Thermodynamics

- Industrial revolution in the late 1700s or early 1800s
- Up till then, heavy work was performed by horses: horsepower
- Fantastic new invention: Heat Engine,
- **Thermodynamics: to gain the maximum efficiency from such heat engines**
- ***Thermodynamics*** deals with the science of "motion" (dynamics) and/or the transformation of "heat" (thermo) and energy into various other energy– containing forms.

Thermodynamic Definitions

Thermodynamic System

- A thermodynamic system is defined as any part of the Universe under consideration.

- It may be something as similar as a beaker of water, a chemical reaction taking place in a test tube, oil refinery or as complicated as an entire galaxy.



Thermodynamic Definitions

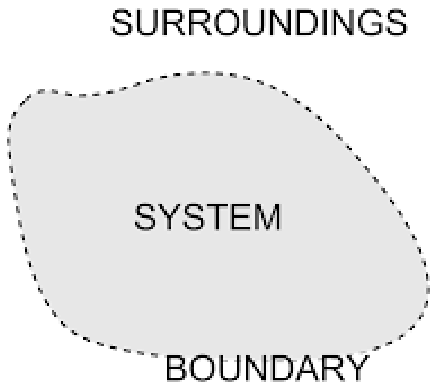
Thermodynamic Surroundings

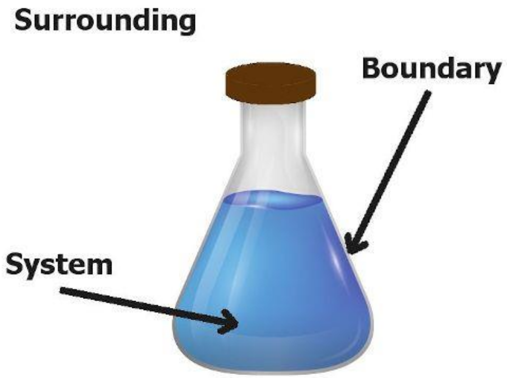
- Thermodynamics surroundings are defined as everything other than the thermodynamic system.

- In other words, the entire rest of the universe.

The Universe

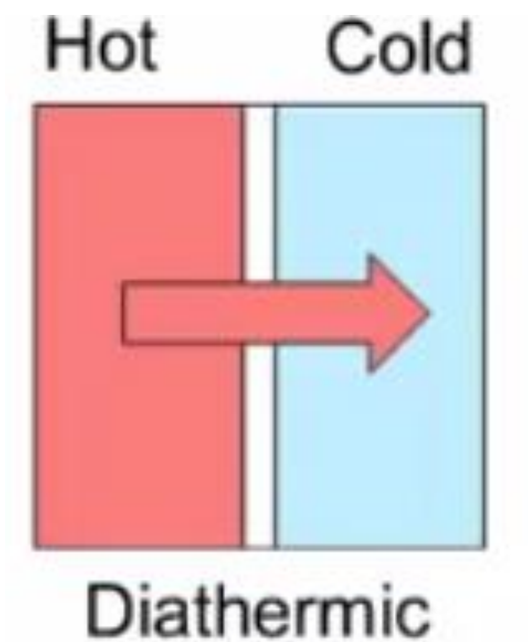
- The universe is therefore the system plus the surroundings.



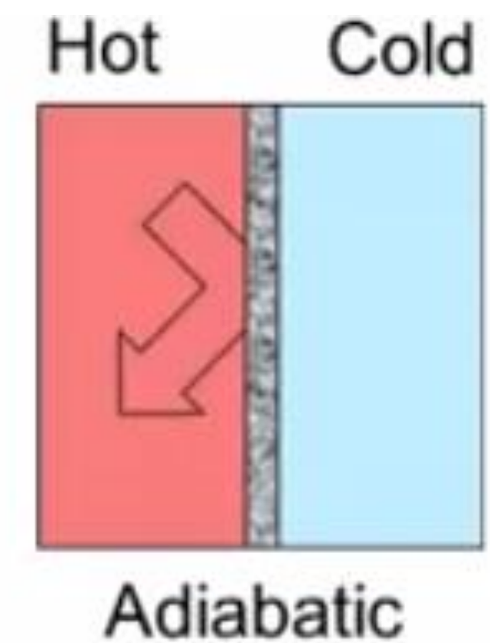


Boundaries

- The boundary may be actual or notional.
- It controls transfer of work, heat, and matter from the system to the surroundings and *vice-versa*.
- The boundary may or may not impose restrictions on such transfers.



A diathermic system allows heat flow into or out of the system.



An adiabatic system prevents heat flow into or out of the system.

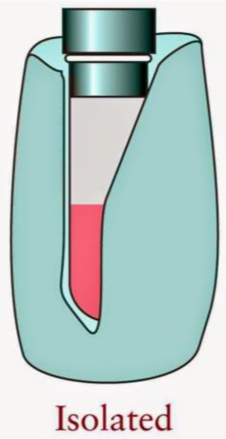
Thermodynamic Definitions

Open, Closed, and Isolated Systems

- **Open System:** An open system may exchange both energy and matter with its surroundings. Changes in composition are therefore possible.

- **Closed System:** A closed system may exchange energy but not matter with its surroundings. Pressure build-up is a distinct possibility.

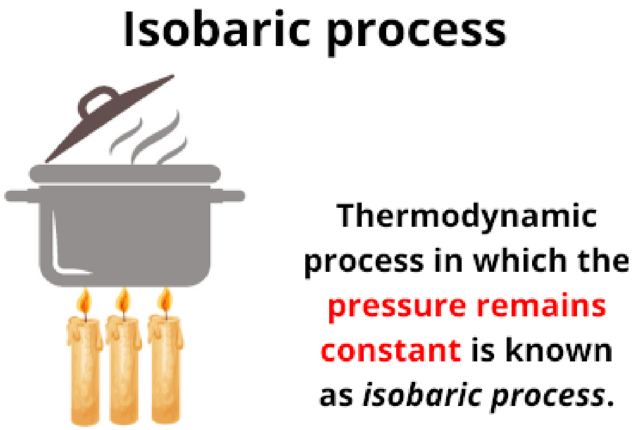
- **Isolated System:** An isolated system may exchange neither energy nor matter with its surroundings. Pressure build-up is a possibility.

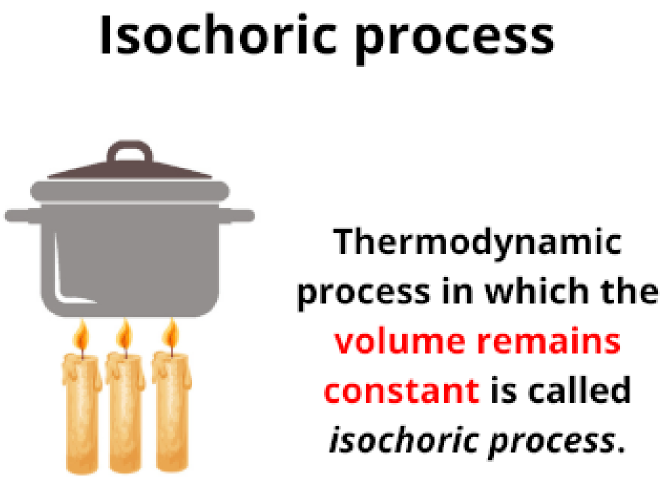


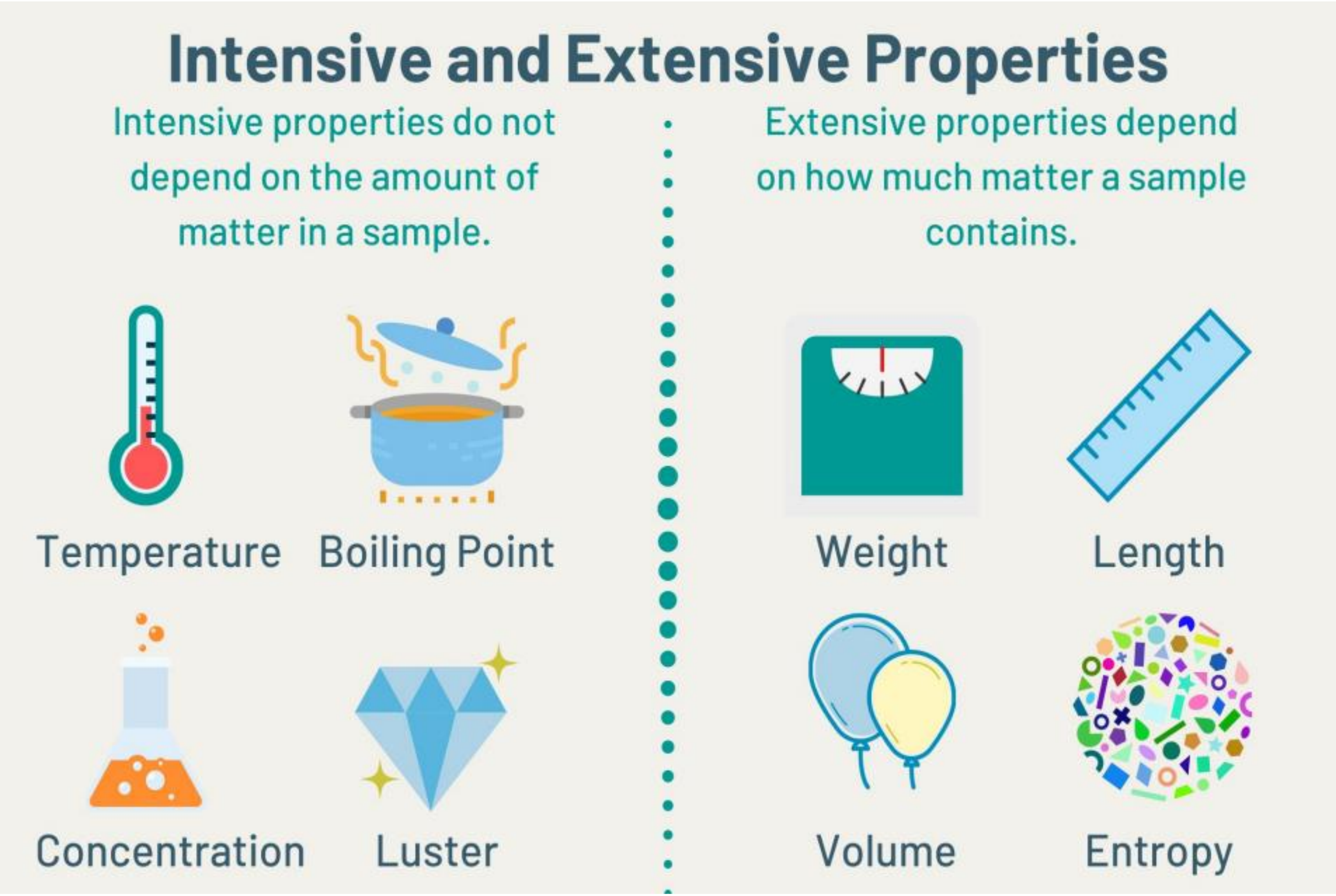
Thermodynamic Definitions

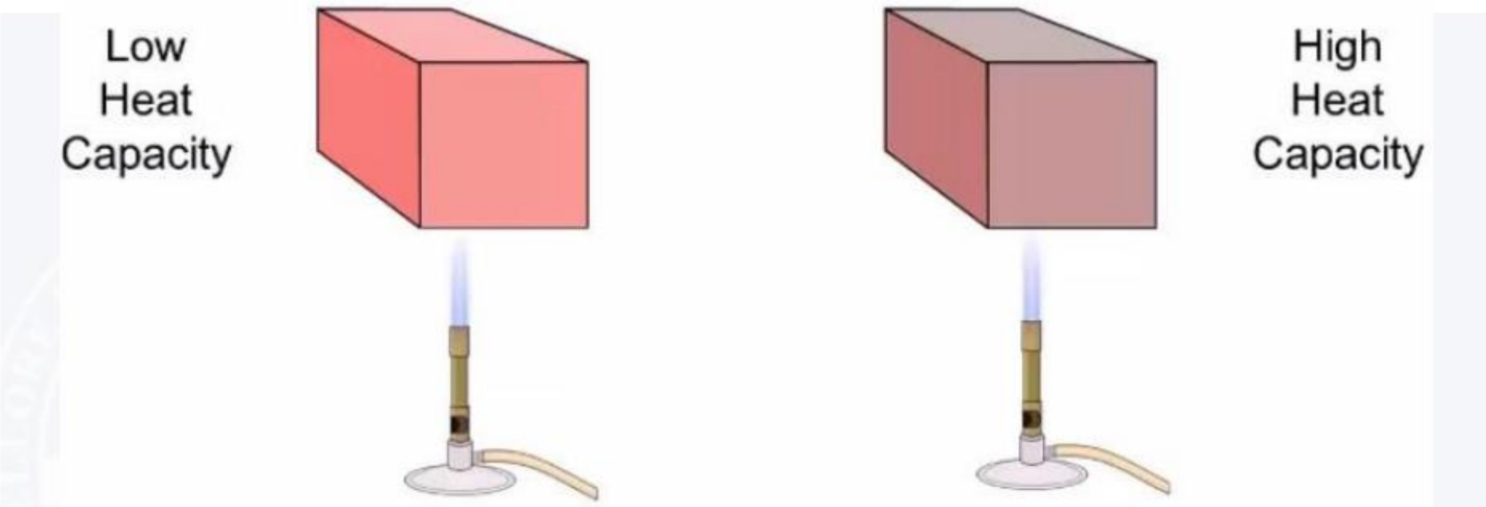
Isothermal, Isobaric, and Isochoric Processes

- Isothermal implies constant temperature, T.
- Isobaric implies constant pressure, P.
- Isochoric implies constant volume, V.






Intensive and Extensive Properties
Intensive properties do not depend on the amount of matter in a sample.
Extensive properties depend on how much matter a sample contains.
Temperature
Boiling Point
Weight
Length
Concentration
Luster
Volume
Entropy



Heat capacity, C, is the heat energy required to raise the temperature of an object by 1° C or 1 K. Measured in J/K.

$$C = q/\Delta T$$

Specific Heat Capacity, $C_s = C / m$ units J K^{-1} kg^{-1}

Molar Heat Capacity, $C_m = C / n$ units J K^{-1} mol^{-1}

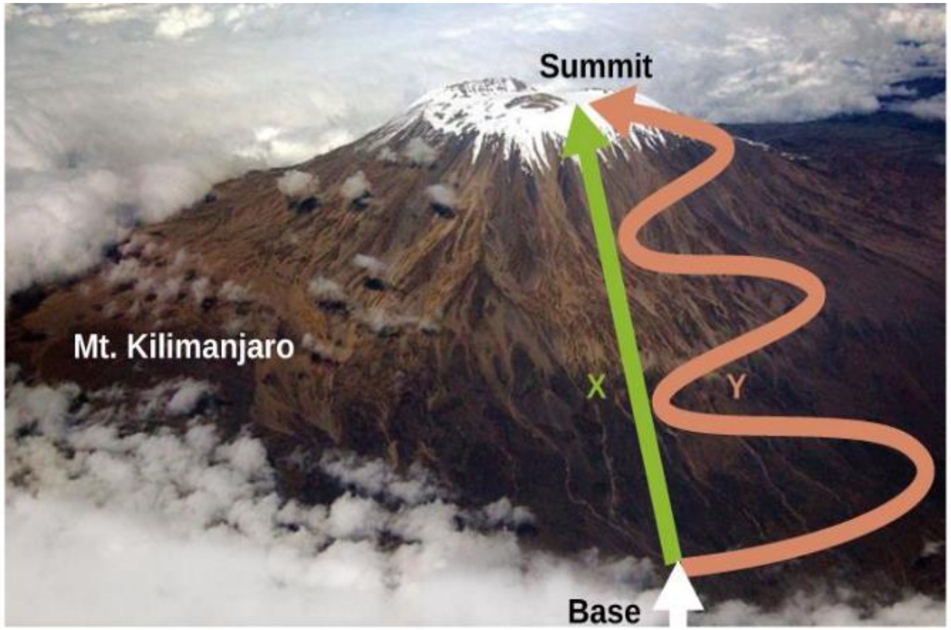
Specific Heat capacity and Molar Heat Capacity are intensive properties.

Thermodynamic Definitions

State Functions

- A state function describes the state of the system. How the system came to be in that state is of no consequence.

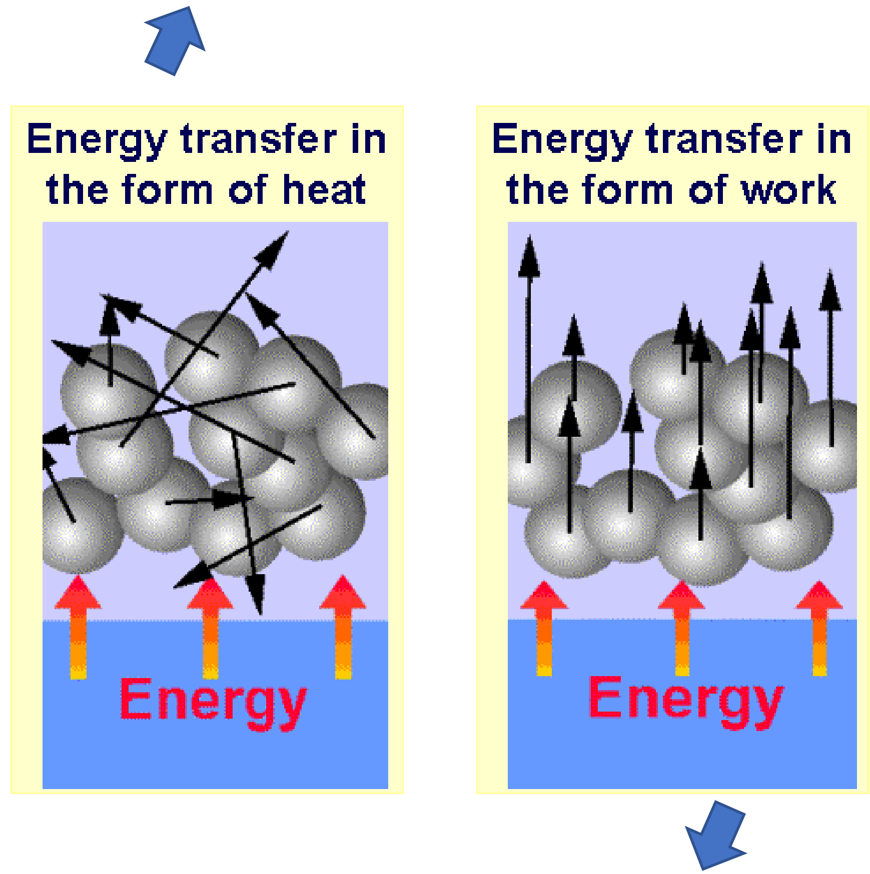
- The following are all state functions:
  Pressure, P;
  Volume, V;
  Temperature, T;
  Mass, m;
  Quantity, n
  Internal Energy, U;
  Enthalpy, H;
  Entropy, S;
  Gibbs Energy, G



Thermodynamic Definitions

Random molecular motion

Path Functions



Uniform molecular motion

- Functions governing transition between the states are called path functions.

- Heat, q & Work, w: are both forms of energy.

- State of a system is changed by supply or removal of energy in the form of heat or work.

Heat and Work Conventions

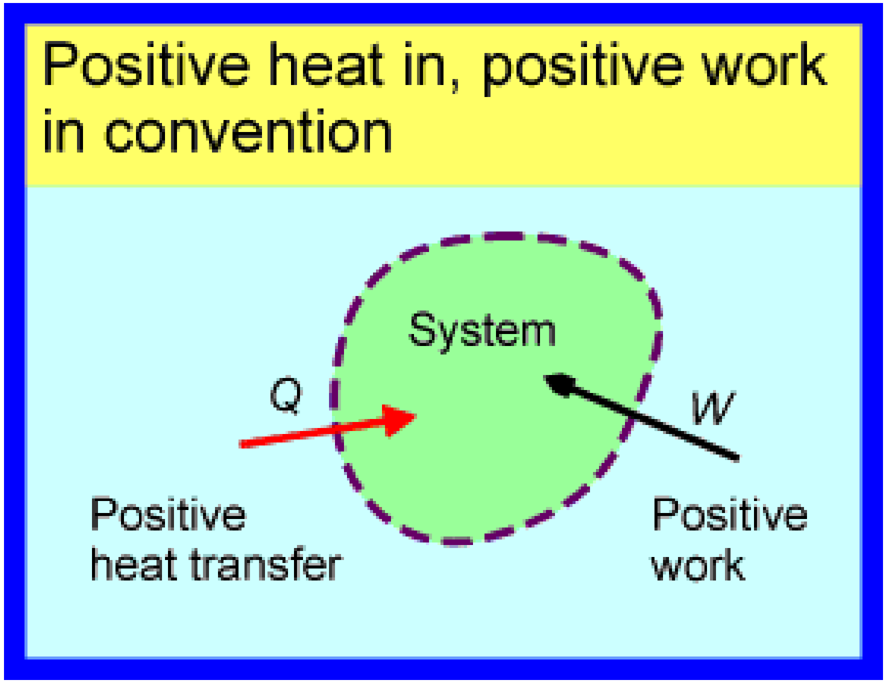
- Path functions are positive when energy enters the system.

- Path functions are negative when energy leaves the system.

- They are often defined as:

  Heat supplied *to* the system, q_{in}

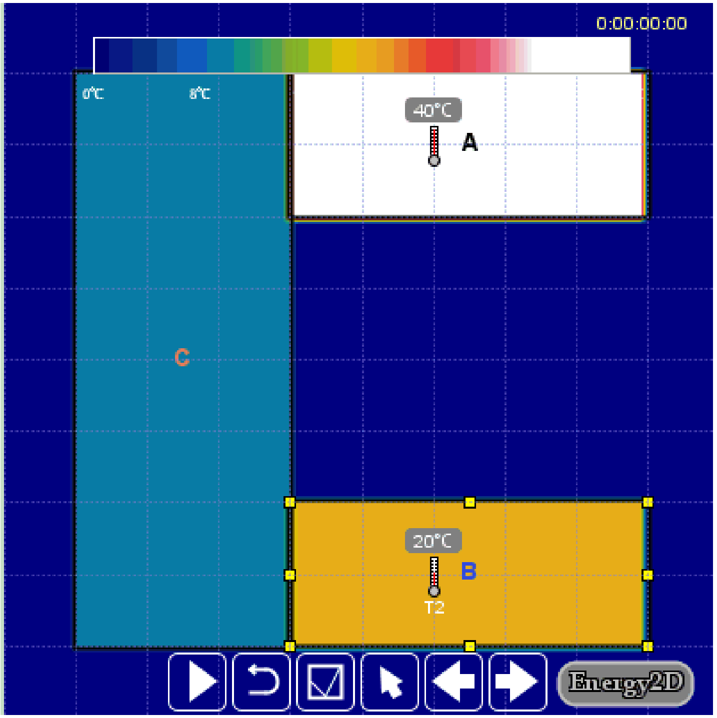
  Mechanical work done *on* the system, W_{on}

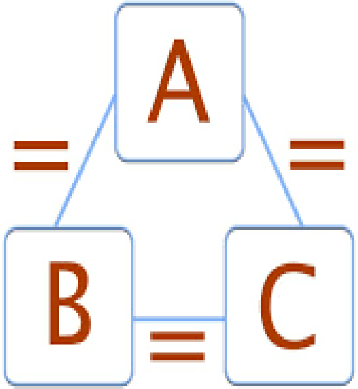


The Zeroth Law of Thermodynamics

- When two objects are separately in thermodynamic equilibrium with a third object, they are in thermodynamic equilibrium with each other.

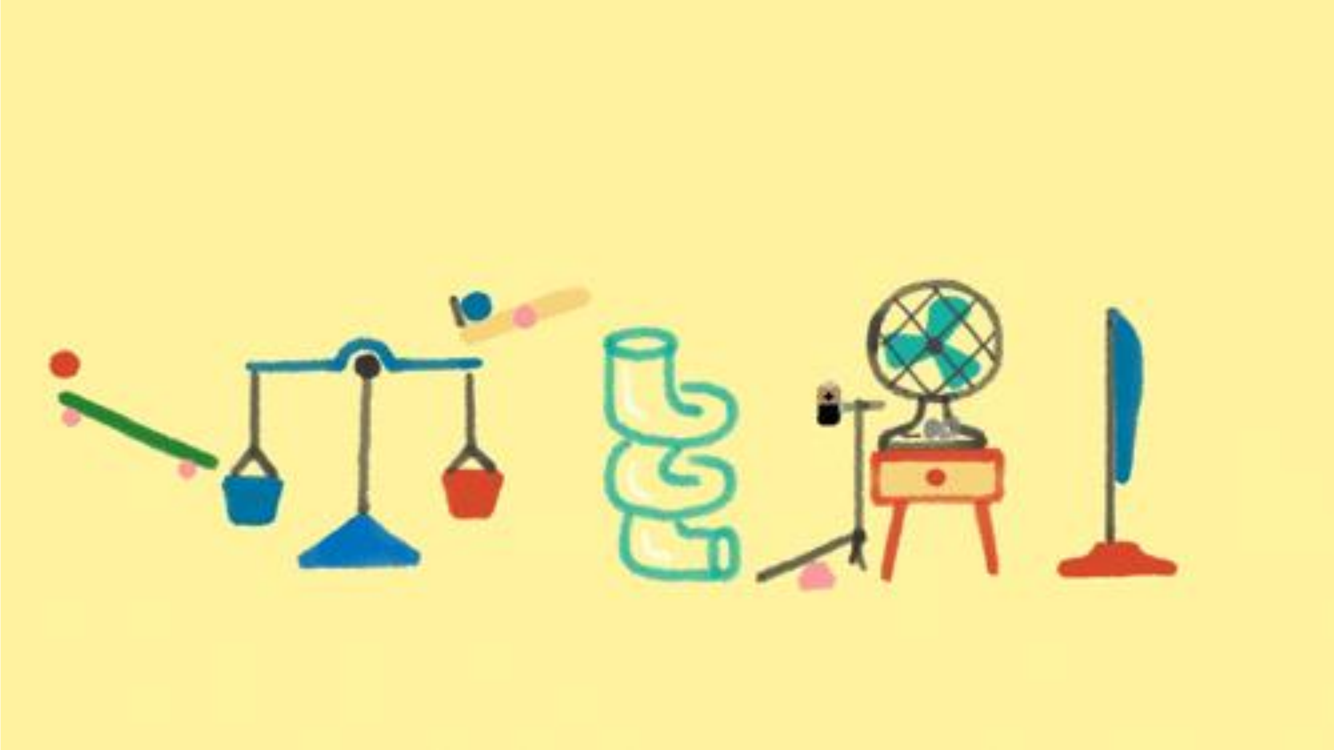
- Whenever two objects are in contact with one another, energy will flow between them until they reach a state of thermodynamic equilibrium. Upon reaching this state we say that the two objects are at the same temperature.





The First Law of Thermodynamics and Enthalpy

- Law of Conservation of Energy: **Energy can neither be created nor destroyed, merely changed from one form to another.**

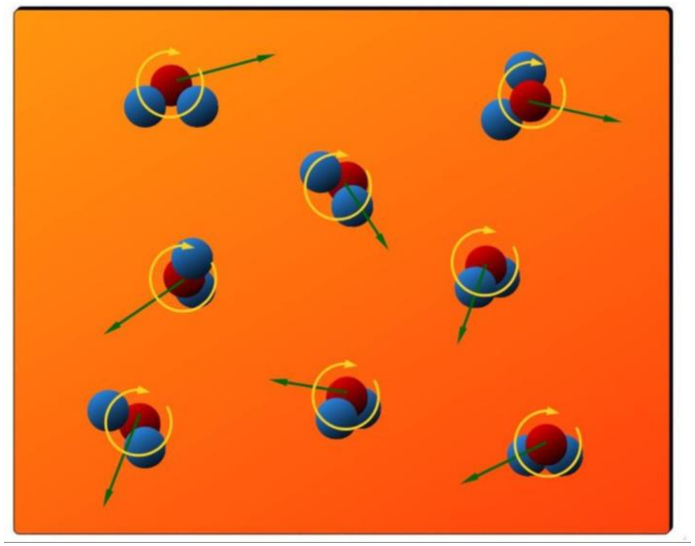


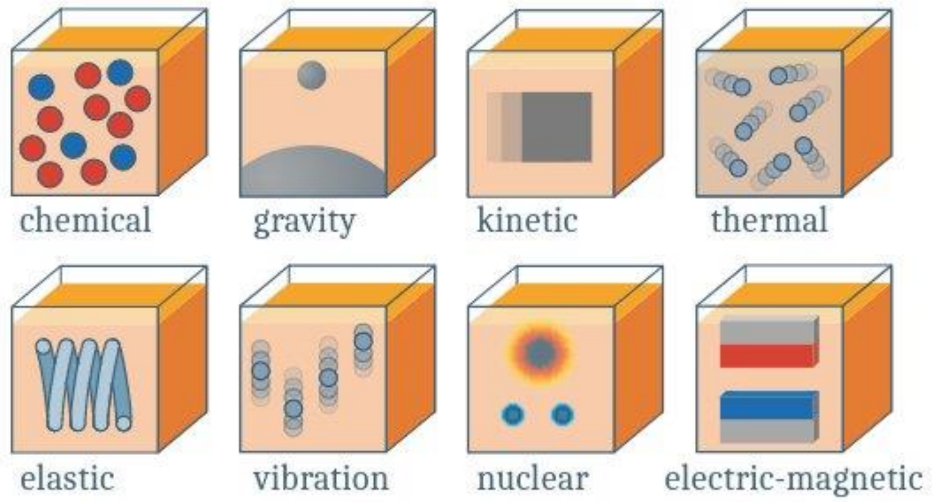
The First Law of Thermodynamics and Enthalpy

- Every thermodynamic system (even all matter) possesses an internal energy, U.

$$\Delta U = q_{in} + w_{on}$$

- Energy is conserved **only for system + surroundings**; system & surroundings can exchange energy





The First Law of Thermodynamics and Enthalpy

- Energy can be transported into a system resulting in an ***increase in internal energy.***
  - The system ***absorbs heat*** from the surroundings: ***+q***
  - or the ***surroundings do work on the system***: ***+w***

Chemical work done during a chemical reaction

Emerging gas performs a chemical work

$$Zn(s) + 2\,HCl(aq) \rightarrow ZnCl_2(aq) + H_2(g)$$

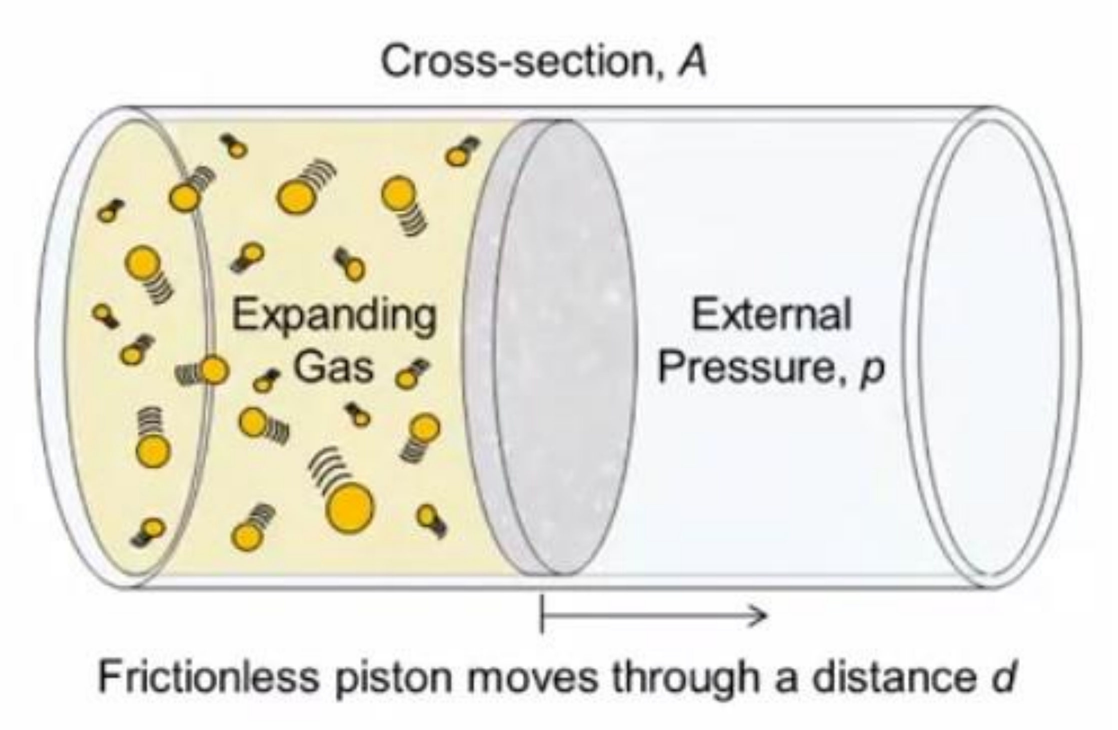
The Work Done by an Expanding gas

$$w = Fd$$

$$p = \frac{F}{A}$$

$$F = pA$$

$$w = pAd$$



$$w = p\Delta V$$

$$w_{on} = -p\Delta V$$

The First Law of Thermodynamics

$$\Delta U = q_{in} + w_{on}$$

$$w_{on} = -p\Delta V$$

$$\Delta U = q_{in} - p\Delta V$$

- Expansion work clearly depends on p and ΔV.

The First Law of Thermodynamics

Free Expansion

$\Delta U = q_{in} - p\Delta V$; but in space p = 0 and $\Delta U = q_{in}$. However, relatively very few experiments are carried out in space.

Reactions at Constant Volume (Domain of Heavy Industries)

$\Delta U = q_{in} - p\Delta V$; but at constant volume ΔV = 0 and $\Delta U = q_v$.
Constant volume reactors must withstand massive pressure changes, heavy chemical industries; very expensive.

Reversible Isothermal Expansion of an Ideal Gas

- **Leads to maximum amount of work performed by the gas.**

Ideal Gas Equation: $pV = nRT$: p = Pressure; V = Volume; T = Absolute temperature measure in K; n = quantity; R = Ideal gas constant, 8.314 J K^{-1} mol^{-1}

$$\Rightarrow \mathrm{p} = \frac{nRT}{V}$$

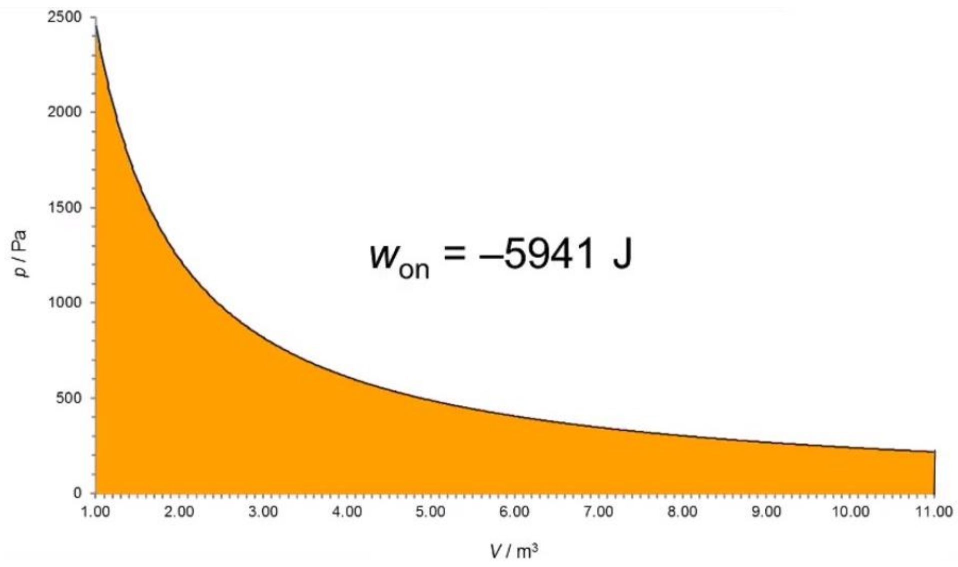
w_{on} for Reversible Isothermal Expansion

$$w_{on} = -pdV$$

But, $$pV = nRT$$

So, $$p = \frac{nRT}{v}$$

$$w_{on} = -\left(\frac{nRT}{v}\right)dV$$



Now consider an expansion from V_i to V_f

$$w_{on} = -nRT\int_{V_i}^{V_f}\frac{dV}{V}$$

$$w_{on} = -nRT\ln\frac{V_f}{V_i}$$

- Reversible change is a theoretical construct.
- Determines maximum possible expansion work.

$$w_{on} = \text{- } 1\text{ mol} \times 8.314\text{ J K}^{-1}\text{ mol}^{-1} \times 298\text{ K} \times \ln(11)$$

The First Law of Thermodynamics

Reactions at Constant Pressure

$\Delta U = q_{in} - p\Delta V$ ⇨ $\Delta U + p\Delta V = q_{in}$ at constant pressure.

- **Now we define a new state function, enthalpy:**

$$H = U + pV$$

$\therefore\ \Delta H = \Delta U + \Delta(pV)$, Use product rule

$\Rightarrow\ \Delta H = \Delta U + p\Delta V + V\Delta p$, but at constant pressure $\Delta p = 0$, $V\Delta p = 0$

$$\therefore \Delta H = \Delta U + p\Delta V$$

⇩

$$\boldsymbol{\Delta H = q_p}$$

Isochoric and Isobaric Heat Capacities

Isochoric heat capacity is defined as:

$$C_V = q_V \,/\, \Delta T = \Delta U \,/\, \Delta T$$

Isobaric heat capacity is defined as:

$$C_p = q_p \,/\, \Delta T = \Delta H \,/\, \Delta T$$

$$C_{p,m} = C_{V,m} + R$$

where $R = 8.314\ \mathrm{J\ K^{-1}\ mol^{-1}}$

The First Law of Thermodynamics Does Not Predict Spontaneous Change

Energy is conserved. It is neither created nor destroyed, but it is transferred in the form of heat and/or work.

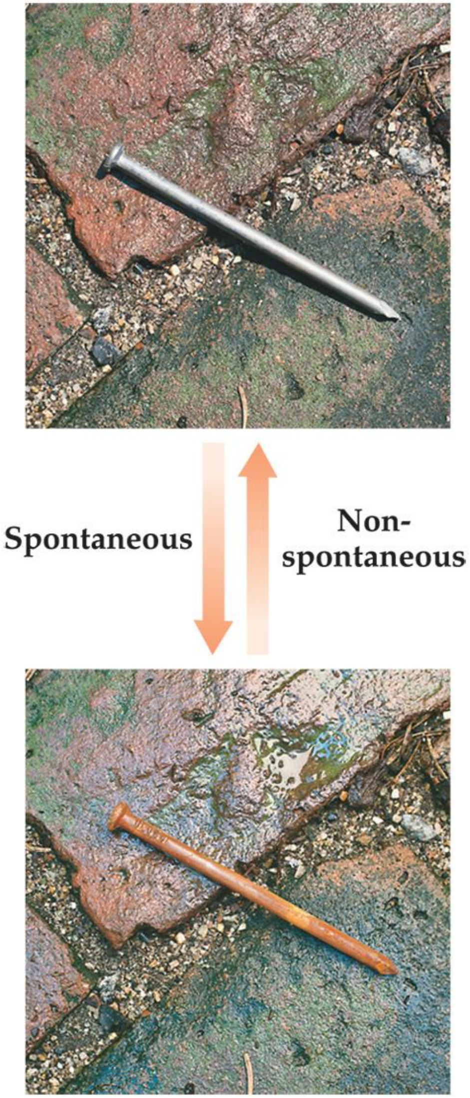
$$\Delta E = q + w$$

The total energy of the universe is constant:

$$\Delta E_{\mathrm{sys}} = -\Delta E_{\mathrm{surr}} \text{ or } \Delta E_{\mathrm{sys}} + \Delta E_{\mathrm{surr}} = \Delta E_{\mathrm{univ}} = 0$$

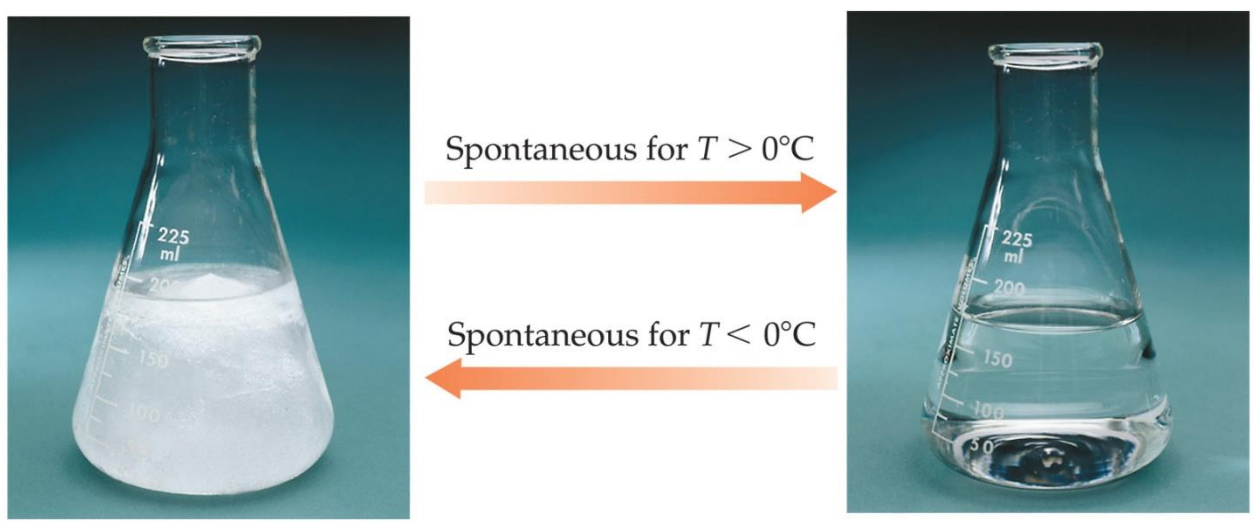
The law of conservation of energy applies to ***all*** changes, but it does ***not*** allow us to predict the ***direction*** of a spontaneous change.

Spontaneous Processes



Processes that are spontaneous in one direction are nonspontaneous in the reverse direction.

Processes that are spontaneous at one temperature may be nonspontaneous at other temperatures.



ΔH Does Not Predict Spontaneous Change

A spontaneous change may be exothermic or endothermic.

Spontaneous ***exothermic*** processes include:

- freezing and condensation at low temperatures,
- combustion reactions,
- oxidation of iron and other metals.

Spontaneous ***endothermic*** processes include:

- melting and vaporization at higher temperatures,
- dissolving of most soluble salts.

The sign of ΔH does not by itself predict the direction of a spontaneous change.

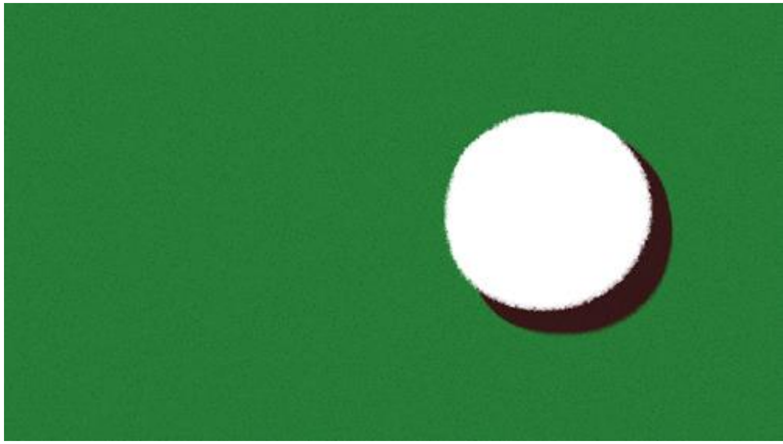
Entropy

- *Entropy* (*S*) is a term coined by Rudolph Clausius in the 19th century.

- Entropy can be thought of as a measure of the randomness of a system.

- It is related to the various modes of motion in molecules.

$$\boxed{\Delta S = \frac{q_{rev}}{T}}$$



The Second Law of Thermodynamics

The sign of ΔS for a reaction does not, by itself, predict the direction of a spontaneous reaction.

If we consider both the system and the surroundings, we find that ***all real processes occur spontaneously in the direction that increases the entropy of the universe.***

For a process to be spontaneous, a ***decrease*** in the entropy of the system must be offset by a ***larger increase*** in the entropy of the surroundings.

$$\boxed{\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} > 0}$$

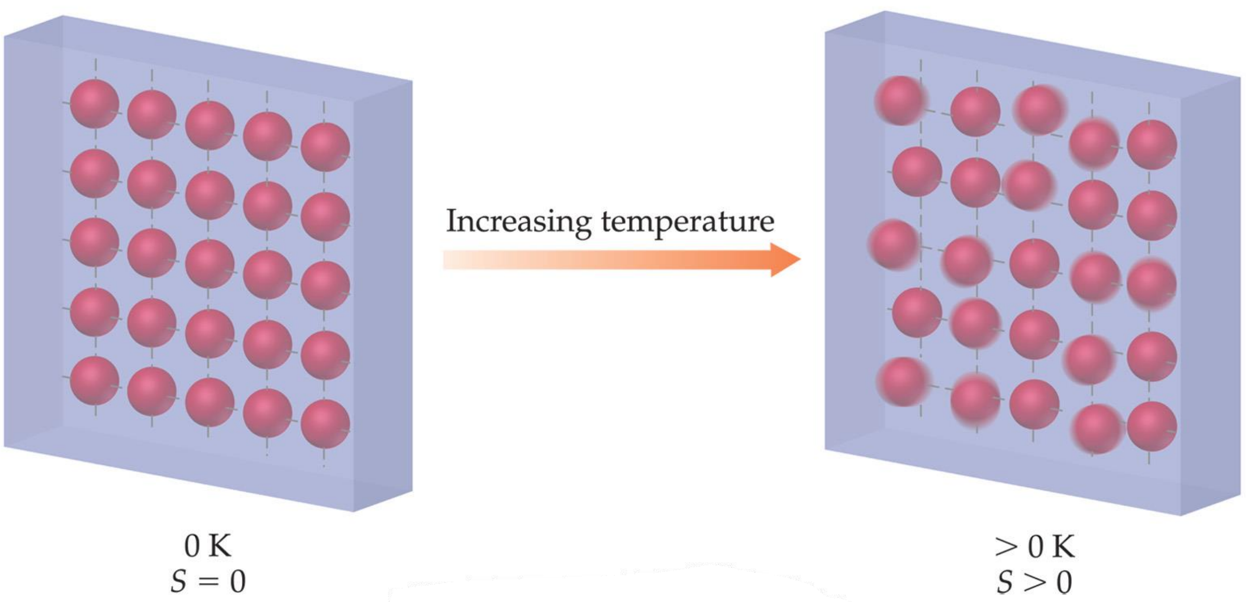
- For enthalpy there is ***no zero point***; we can only measure ***changes*** in enthalpy.
- For entropy there ***is*** a zero point, and we can determine ***absolute*** entropy values.

The Third Law of Thermodynamics

A perfect crystal has zero entropy at absolute zero.

$$\boxed{S_{sys} = 0 \text{ at } 0\text{ K}}$$

A "perfect" crystal has flawless alignment of all its particles. At absolute zero, the particles have minimum energy, so there is only one microstate.



Standard Molar Entropies

$S°$ is the ***standard molar entropy*** of a substance, measured for a substance in its ***standard state*** in units of J/mol·K.

The conventions for defining a standard state include:

- 1 atm for gases
- 1 *M* for solutions, and
- the pure substance in its most stable form for solids and liquids.

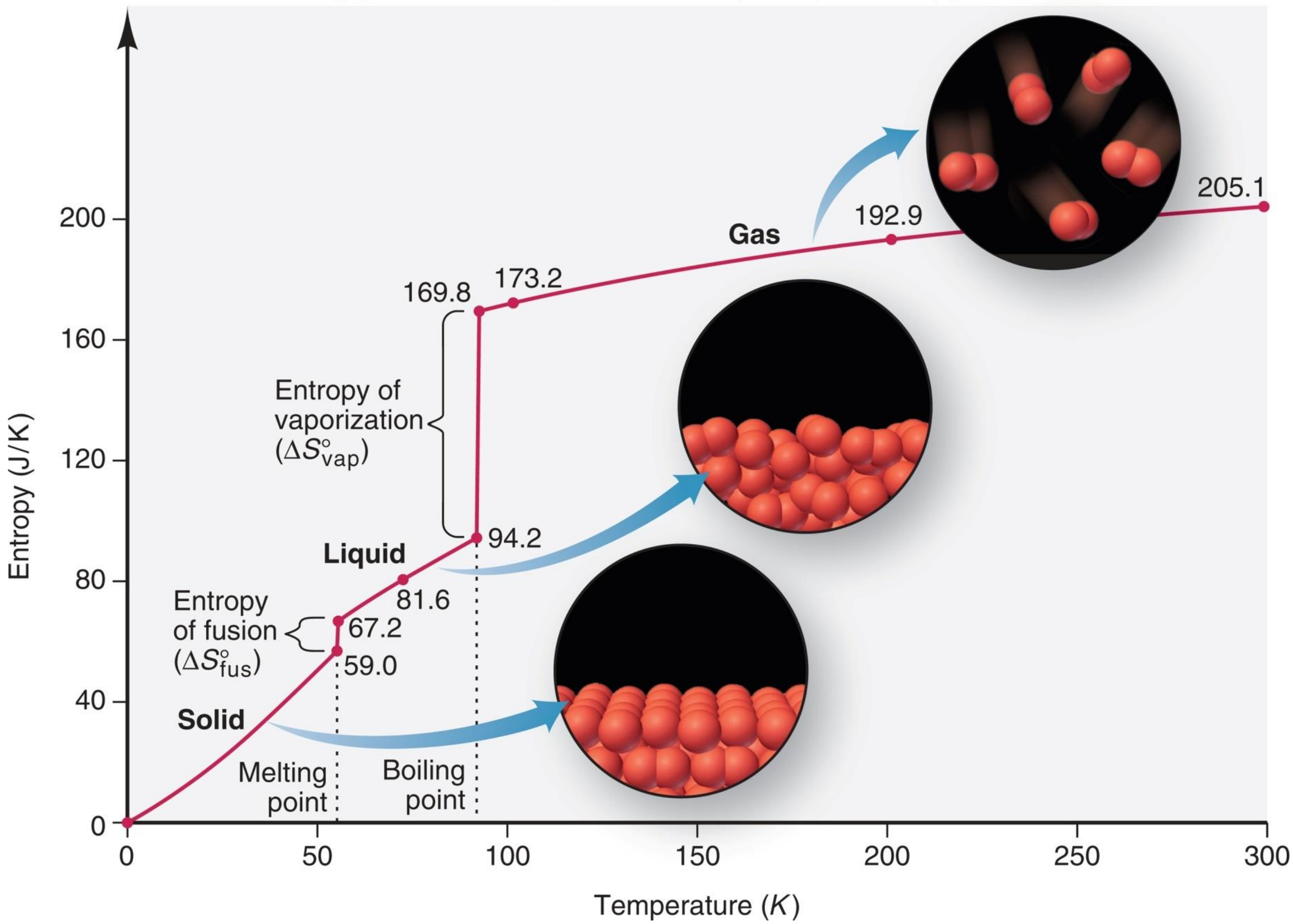
TABLE 19.2 Standard Molar Entropies of Selected Substances at 298 K

Substance	$S°$, J/mol-K
Gases	
$H_2(g)$	130.6
$N_2(g)$	191.5
$O_2(g)$	205.0
$H_2O(g)$	188.8
$NH_3(g)$	192.5
$CH_3OH(g)$	237.6
$C_6H_6(g)$	269.2
Liquids	
$H_2O(l)$	69.9
$CH_3OH(l)$	126.8
$C_6H_6(l)$	172.8
Solids	
$Li(s)$	29.1
$Na(s)$	51.4
$K(s)$	64.7
$Fe(s)$	27.23
$FeCl_3(s)$	142.3
$NaCl(s)$	72.3

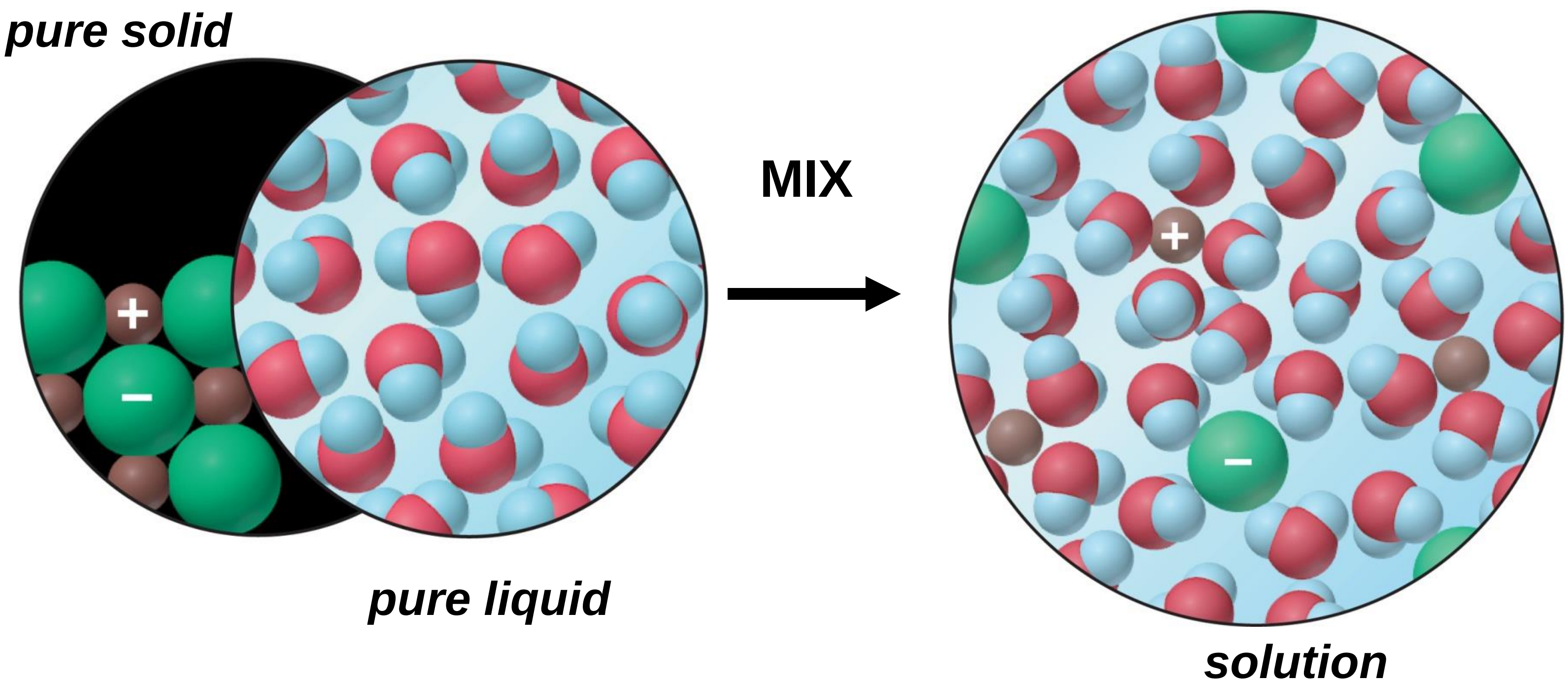
Factors Affecting Entropy

- Entropy depends on temperature.
  - For any substance, $S°$ increases as temperature increases.
- Entropy depends on the physical state of a substance.
  - $S°$ increases as the phase changes from solid to liquid to gas.
- The formation of a solution affects entropy.
- Entropy is related to atomic size and molecular complexity.
  - Remember to compare substances in the same physical state.

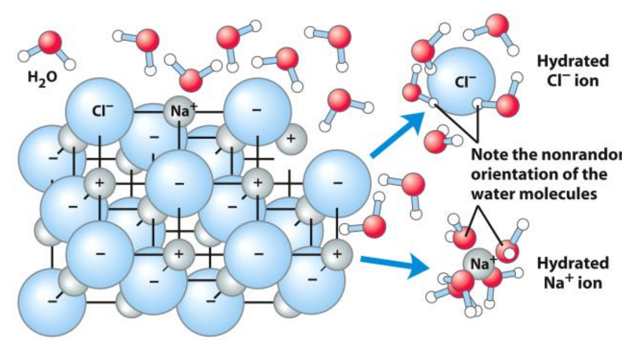
The increase in entropy during phase changes from solid to liquid to gas.



The entropy change accompanying the dissolution of a salt.



The entropy of a salt solution is usually ***greater*** than that of the solid and of water, but it is affected by the organization of the water molecules around each ion.

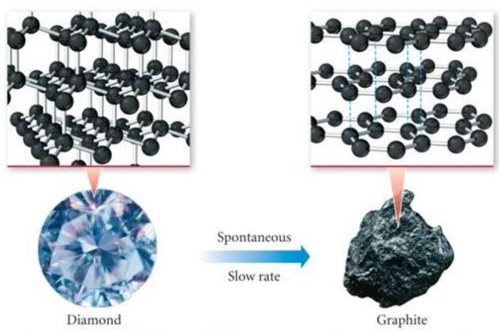


Entropy and Structure

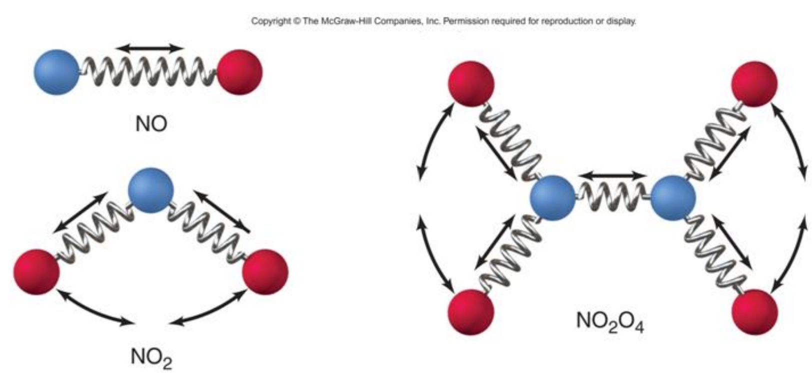
For allotropes, S° is higher in the form that allows the atoms more freedom of motion.

S° of graphite is 5.69 J/mol·K, whereas S° of diamond is 2.44 J/mol·K.

Diamond → Graphite



Graphite has less FE than diamond, so the conversion of diamond into graphite is spontaneous – but don't worry, it's so slow that your ring won't turn into pencil lead in your lifetime (or through many of your generations).



Entropy Changes in the System

The ***standard entropy of reaction***, ΔS°_{rxn}, is the entropy change that occurs when all reactants and products are in their standard states.

$$\boxed{\Delta S^\circ_{rxn} = \Sigma m S^\circ_{products} - \Sigma n S^\circ_{reactants}}$$

where *m* and *n* are the amounts (mol) of products and reactants, given by the coefficients in the balanced equation.

Calculating the Standard Entropy of Reaction, ΔS°_{rxn}

PROBLEM: Predict the sign of ΔS°_{rxn} and calculate its value for the combustion of 1 mol of propane at 25° C.

$$C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(l)$$

PLAN: From the change in the number of moles of gas (6 mol yields 3 mol), the entropy should decrease ($\Delta S^\circ_{rxn} < 0$).

SOLUTION:

$$\Delta S^\circ_{rxn} = [(3\text{ mol } CO_2)(S^\circ \text{ of } CO_2) + (4\text{ mol } H_2O)(S^\circ \text{ of } H_2O)] - [(1\text{ mol } C_3H_8)(S^\circ \text{ of } C_3H_8) + (5\text{ mol } O_2)(S^\circ \text{ of } O_2)$$

$$= [(3\text{mol})(213.7\text{J/K}\cdot\text{mol}) + (4\text{ mol})(69.9\text{J/K}\cdot\text{mol})] - [(1\text{mol})(269.9\text{ J/K}\cdot\text{mol}) + (5\text{ mol})(205.0\text{ J/K}\cdot\text{mol})$$

$$\boxed{= -374\text{ J/K}}$$

Entropy Changes in the Surroundings

A ***decrease*** in the entropy of the system is ***outweighed*** by an ***increase*** in the entropy of the surroundings.

The surroundings function as a heat source or heat sink.

In an ***exothermic*** process, the surroundings absorbs the heat released by the system, and S_{surr} increases.

$$q_{sys} < 0; \;\; q_{surr} > 0 \text{ and } \Delta S_{surr} > 0$$

In an ***endothermic*** process, the surroundings provides the heat absorbed by the system, and S_{surr} decreases.

$$q_{sys} > 0; \;\; q_{surr} < 0 \text{ and } \Delta S_{surr} < 0$$

Temperature at which Heat is Transferred

For any reaction, $q_{sys} = -q_{surr,}$ the heat transferred is specific for the reaction and is the ***same*** regardless of the temperature of the surroundings.

$$\Delta S_{surr} = -\frac{q_{sys}}{T}$$

$$\boxed{\boldsymbol{\Delta S_{surr} = -\frac{\Delta H_{sys}}{T}}}$$ for a process at constant *P*.

Surroundings & system

$$\Delta S_{universe} = \Delta S_{system} + \Delta S_{surround}$$

$$\Delta S_{surr} = \frac{-\Delta H^\circ_{sys}}{T}$$

$$\Delta S_{universe} = \Delta S_{system} + \frac{-\Delta H^\circ_{sys}}{T}$$

$$\underbrace{T\Delta S_{universe}} = T\Delta S_{system} + -\Delta H^\circ_{sys}$$

= – Gibbs Free Energy

$$\underbrace{T\Delta S_{universe}} = T\Delta S_{system} + -\Delta H^{\circ}_{sys}$$

= – Gibbs Free Energy

Make this equation nicer:

$$-T\Delta S_{universe} = \Delta H^{\circ}_{sys} - T\Delta S_{system}$$

$$\boxed{\Delta G = \Delta H^{\circ}_{sys} - T\Delta S_{system}}$$

Gibbs Free Energy

The Gibbs free energy (G) combines the enthalpy and entropy of a system;
$G = H - TS$.

The ***free energy change*** (ΔG) is a measure of the spontaneity of a process and of the useful energy available from it.

$$\boxed{\Delta G_{sys} = \Delta H_{sys} - T\,\Delta S_{sys}}$$

$\Delta G < 0$ for a spontaneous process

$\Delta G > 0$ for a nonspontaneous process

$\Delta G = 0$ for a process at equilibrium

Calculating $\Delta G°$

$\Delta G°_{rxn}$ can be calculated using the Gibbs equation:

$$\boxed{\Delta G°_{sys} = \Delta H°_{sys} - T\,\Delta S°_{sys}}$$

$\Delta G°_{rxn}$ can also be calculated using values for the ***standard free energy of formation*** of the components.

$$\boxed{\Delta G°_{rxn} = \Sigma m\,\Delta G°_{products} - \Sigma n\,\Delta G°_{reactants}}$$

Reaction Spontaneity and the Signs of ΔH, ΔS, and ΔG

ΔH	ΔS	$-T\Delta S$	ΔG	**Description**
–	+	–	–	Spontaneous at all *T*
+	–	+	+	Nonspontaneous at all *T*
+	+	–	+ or –	Spontaneous at higher *T;* nonspontaneous at lower *T*
–	–	+	+ or –	Spontaneous at lower *T*; nonspontaneous at higher *T*

Free Energy and Equilibrium

$$\boxed{\Delta G = RT \ln\frac{Q}{K} = RT \ln Q - RT \ln K}$$

If Q and K are very different, ΔG has a very large value (positive or negative). The reaction releases or absorbs a large amount of free energy.

If Q and K are nearly the same, ΔG has a very small value (positive or negative). The reaction releases or absorbs very little free energy.

For standard state conditions, $Q = 1$ and $\boxed{\Delta G^\circ = -RT \ln K}$

To calculate ΔG for any conditions:

$$\boxed{\Delta G = \Delta G^\circ + RT \ln Q}$$